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Palaeomagnetic results from the Permian Saint-Affrique basin (France) and implications for late and post-Hercynian tectonics

A. Diego-Orozco [a,b] and B. Henry [*,a]

[a] *Laboratoire de Géomagnétisme, CNRS and Université Paris VI, 4 avenue de Neptune, 94107 Saint-Maur Cedex, France*
[b] *Instituto Mexicano del Petróleo, Eje central Lázuro Cárdenas 152, 07730 Mexico City, Mexico*

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ABSTRACT

Palaeomagnetic analyses have been carried out in sedimentary formations of the Permian Saint-Affrique basin of France and indicate a late Saxonian–early Thuringian age (the Dourdou sandstones) and at the top of the succession a middle Thuringian age (the Saint-Pierre pelites). Important rotations, clockwise and anticlockwise, respectively, on the northern and southern borders of the basin, affected the Saint-Affrique area between the late Autunian and Saxonian. The opening of the basin occurred without rotation from the Saxonian to the middle Thuringian. Comparison with corresponding results from stable Europe, and in particular from the neighbouring Lodève basin, indicates an anticlockwise rotation of the entire Saint-Affrique area of the order of 7° after the middle Thuringian (Palatine phase or a more recent tectonic event).

Introduction

The Saint-Affrique basin is filled by an important succession, beginning with strata of upper Autunian age. The main aim in this palaeomagnetic study was to search for traces of rotations that may have occurred and to determine their importance and age. Indeed palaeomagnetism appears to be one of the most useful methods for structural analysis, and can be applied to study the late and post-Hercynian tectonic evolution of this area.

Van Den Driessche and Brun (1989) assume a reactivation of Hercynian thrusting under the Saint-Affrique basin during late Hercynian extension (Santouil, 1980), implying possible rotations of this area; evidence of such rotations has been obtained locally on the southern border of the basin (Cogné et al., 1990). Sedimentological observations (Legrand, 1990) have pointed to displacement along listric faults during the deposition of the Permian succession.

Geological setting

The Permian Saint-Affrique basin (Fig. 1) is situated south of the French Massif Central (42.9°N 2.8°E). It is limited to the north by the Rouergue basement (Lévezou dome), to the west by the metamorphic Albigeois thrust sheets, to the south by the Hercynian Montagne Noire (Lacaune) and to the east by the Mesozoic cover of the Causses. It appears as an E–W half-graben (Fig. 2), with fault systems mainly along its southern margin, filled by conglomerates, sandstones and pelites (generally with a gentle southward dip) overlying Hercynian basement or, locally, Stephanian detrital sediments. Microtectonic data show the opening of the basin as the result of an extension, oriented NE–SW, during the late Palaeozoic (Santouil, 1980). This orientation is roughly parallel to the transfer faults (Plaisance and Cévennes faults) of the extensional system between the Saint-Affrique basin and the Montagne Noire; the l'Espinouse fault between these transfer faults likely corresponds to a Hercynian deep flat-lying detachment, under the Saint-Affrique area, reactivated during this extension (Van Den Driessche and Brun, 1989). Such an assump-

* Corresponding author.

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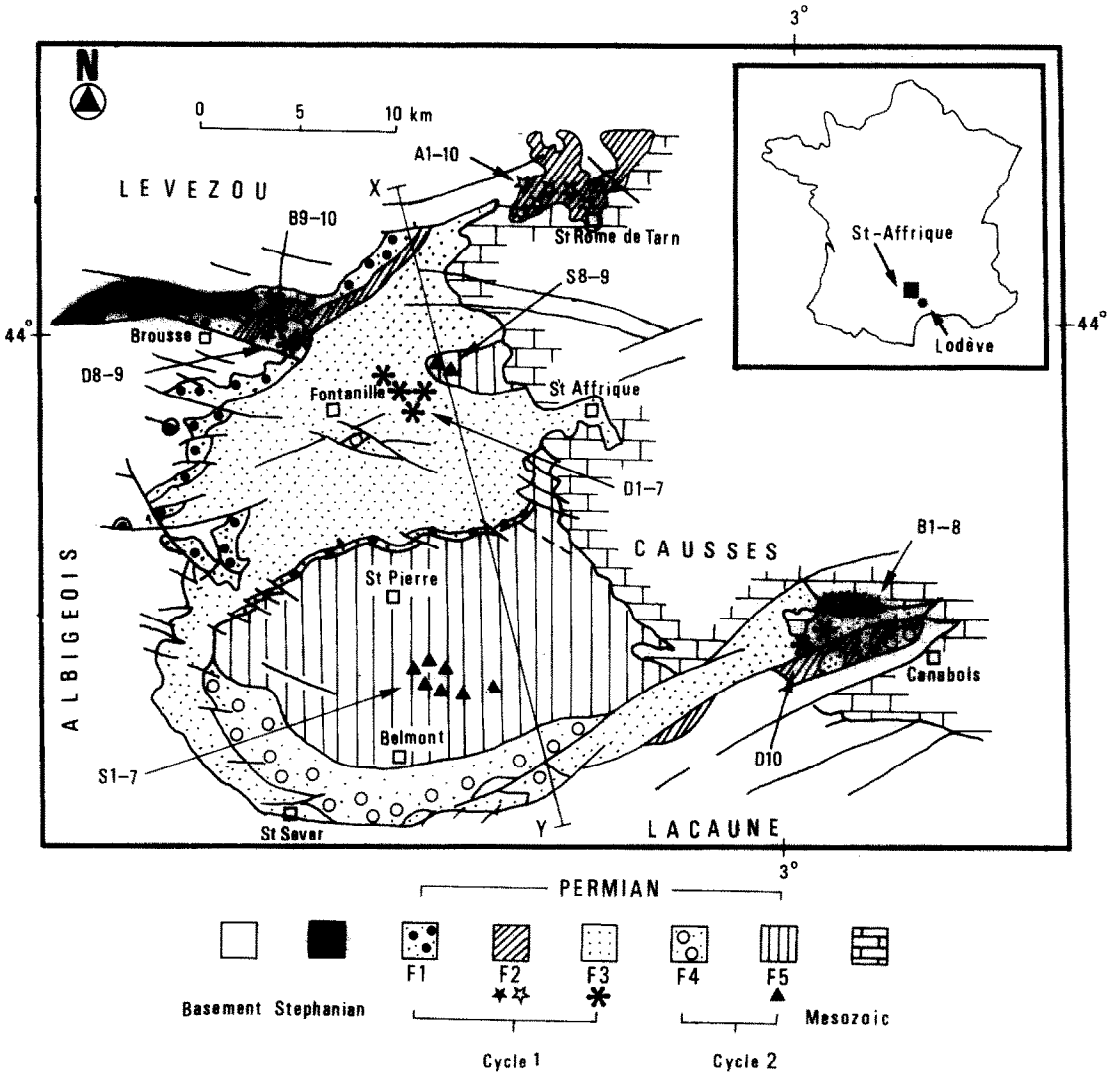


Fig. 1. Geological map (after Rolando, 1988) and sampling sites: Saint-Rome 2a (white stars, sites A1–10); Saint-Rome 2b (black stars, sites B1–10); Dourdou (black flower; sites D1–10) and Saint-Pierre (triangles, sites S1–9).

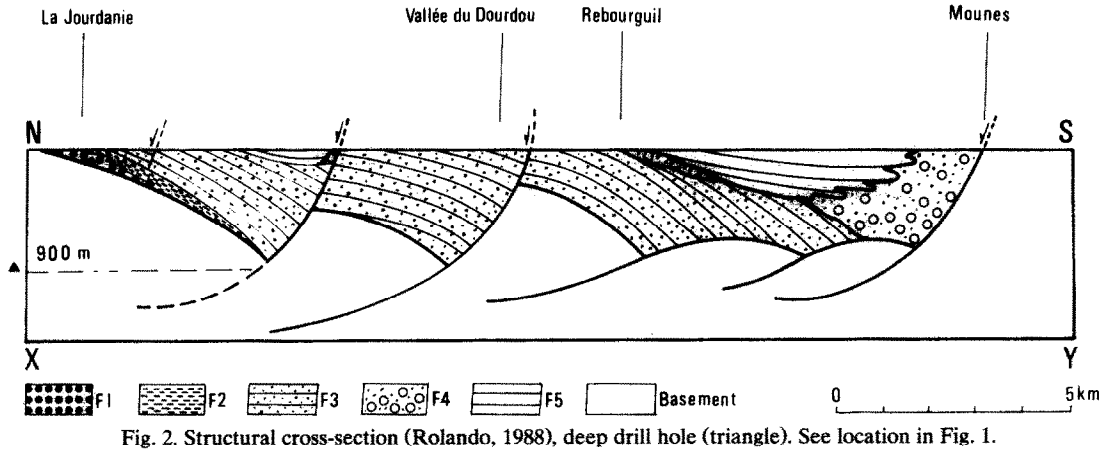


Fig. 2. Structural cross-section (Rolando, 1988), deep drill hole (triangle). See location in Fig. 1.

tion is in agreement with the tectono-sedimentological evolution (roll-over structures at all scales) of the southern margin of the basin (Legrand, 1990; Legrand et al., 1991a,b). According to Legrand (1990), deformations (N–S compression, followed by occurrence of N–S normal faults) later affected this area before a N–S extension (Santouil, 1980), which should correspond to dextral movements along deep structures (likely ENE–WSW, Legrand, 1990) and to the location of barite veins in the northern border of the basin; effects of this N–S extension are also visible in the Mesozoic cover. The Saint-Affrique basin was affected later by the N–S Pyrenean compression (Santouil, 1980).

The Permian sediments include five detrital continental formations (Fig. 3), called F1 to F5 (Rolando, 1988). Palynological associations yield dates for part of the series (Rolando et al., 1989): F1 and F2a formations are of upper Autunian age, and F3 of lower Thuringian (Rolando, 1988), possibly late Saxonian (Schaarschmidt, 1980) ages. F2b is attributed to the lower Saxonian, because of its stratigraphical position and by correlation of sequential evolution and of cinerite levels with the Lodève basin (Doubinger et al., 1987; Rolando

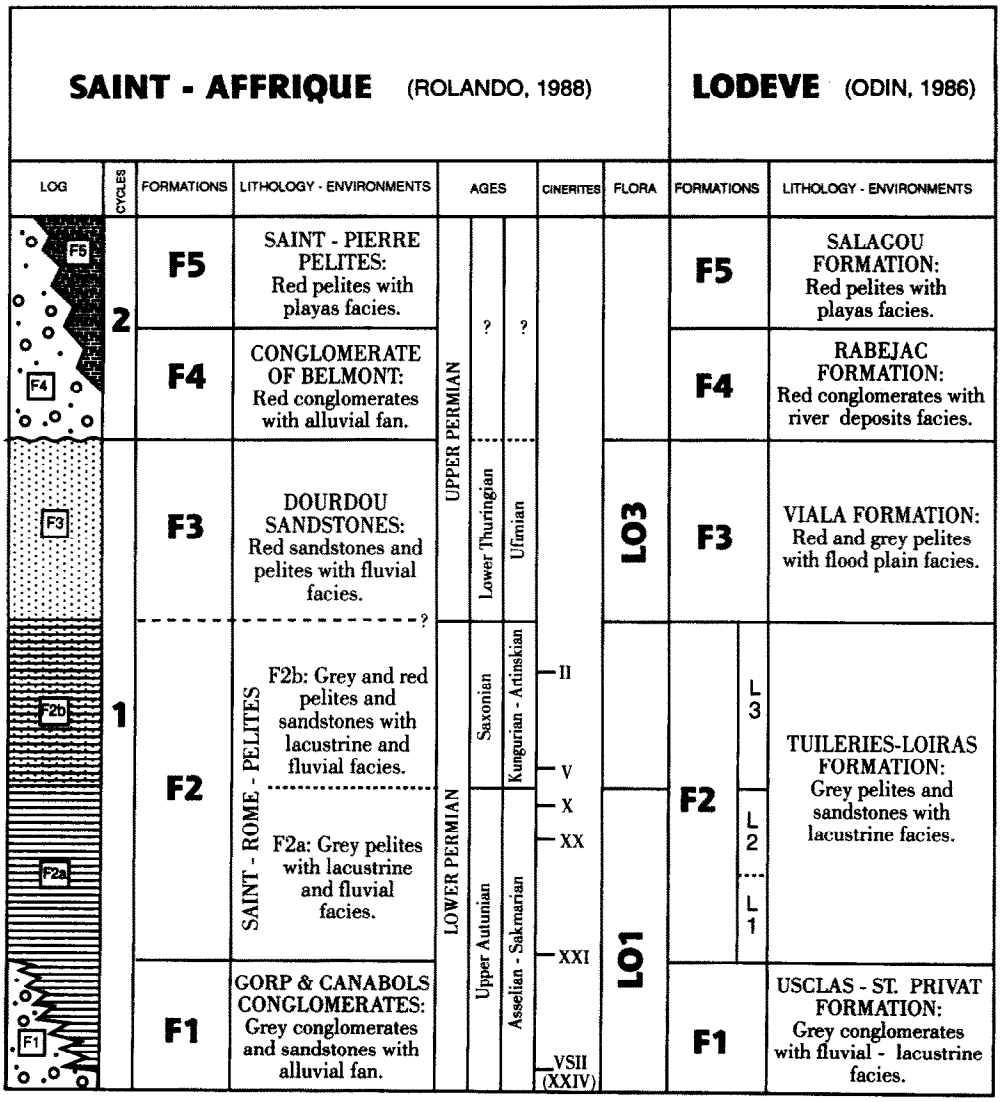


Fig. 3. Stratigraphical series of Saint-Affrique and correlations (after Rolando, 1988) with the Lodève basin (Odin, 1986).

et al., 1989). F5 levels are undated (overlying formations are attributed to the Middle Triassic); K–Ar dating (Bonhomme et al., 1980) yields, however, a middle Thuringian age (about 250 Ma) for diagenesis phenomena attributed to deposition from these F5 levels.

Only the finest-grained facies were selected for this study: *Saint-Rome pelites (F2)*, composed of grey pelites with lacustrine facies (F2a), and grey and red pelites with lacustrine and fluvial facies (F2b); *Dourdou sandstones (F3)*, made up of red pelites to sandstones with fluvial facies; and *Saint-Pierre pelites (F5)*, composed of red pelites with playas facies.

Sampling and magnetic measurement procedures

115 handsamples (using horizontal plaster caps with magnetic and sun sightings for orientation—Thellier, 1938) were collected all over the basin at 39 sites. Most of the formations were sampled at different areas to have a fold test, or to point out eventual local rotations.

At least three standard cylinder specimens were drilled and cut in the laboratory from each sample. Prior to demagnetization analysis, they were stored in a zero-magnetic field for at least one month to allow for a decay of the possible viscous remanent magnetization (VRM) acquired in situ or after sampling. The NRM was measured by a JR-4 spinner magnetometer (Geofyzika, Brno). Thermal demagnetization was carried out using a cylindrical furnace, with a fast heating level, and with cooling by forced air at the heating site. Reduced temperature intervals were used at the highest temperatures to better determine the remanence evolution; selected thermal steps were 100° up to 200°C treatment, and 50° to 20°C for higher temperatures.

Alternating field demagnetization up to 160 mT was performed using an apparatus cancelling automatically the parasitic anhysteretic effect (Le Goff, 1985); the steps used for this treatment were of 5 mT to 20 mT. Fisher (1953) and bivariate statistics (Le Goff, 1990; Le Goff et al., 1992) providing a confidence ellipse around a vectorial weighted mean, were used for data analyses.

Rock magnetism study and palaeomagnetic measurements

The natural remanent magnetization (NRM) of 230 specimens was first measured. Two main different groups of magnetization intensities can be distinguished (Fig. 4). The first one is grouping intensities ranging between 10 and 500×10^{-4} A/m from pelitic grey formation (Saint-Rome 2a and locally 2b); the second group shows NRM intensities between 500 and 6000×10^{-4} A/m from the red sandstones (Dourdou), red and grey

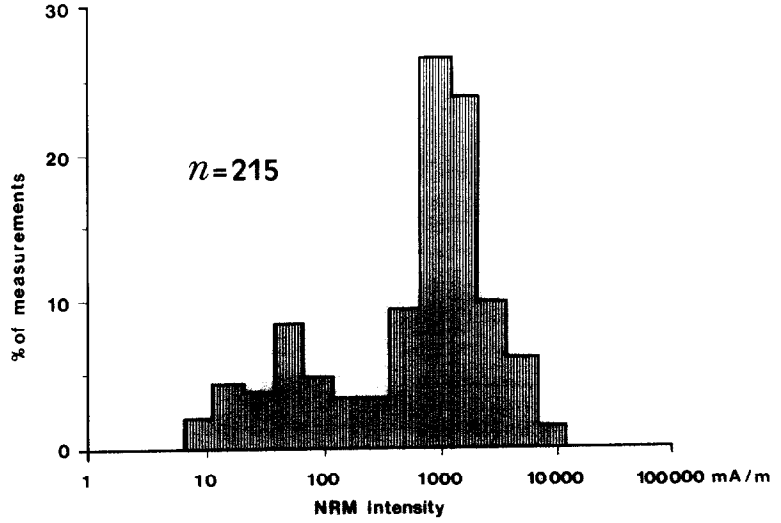


Fig. 4. NRM intensity histogram.

pelites (Saint-Rome 2b) and red pelites (Saint-Pierre). NRM intensities are mostly between 1000 and 3000×10^{-4} A/m, but sometimes reach values as low as 200 (grey levels of Saint-Rome 2b) and as high as 6000×10^{-4} A/m (Saint-Pierre). This wide distribution of NRM intensities can be interpreted, for magnetic carriers, as a variation of concentration as well as mineralogical difference. As facies in the Saint-Affrique basin are very similar to those of Lodève basin, it can be noticed that intensity values are similar in these basins for grey pelites (magnetite identified in the Lodève and Saint-Affrique basins: Mérabet and Guillaume, 1988, and Cogné et al., 1990) and grey and red levels (magnetite and hematite recognized in Lodève: Mérabet and Guillaume, 1988; Cogné et al., 1990).

To specify the magnetic phases present in the Saint-Affrique basin, hysteresis loops of samples were determined, using a translation inductometer in an electromagnet providing a field up to 1.6 T. All the obtained loops are almost perfectly linear, indicating the strongly dominant effect of the paramagnetic minerals; however, some data concerning the ferrimagnetics can be obtained from results of grey and red samples.

(a) In grey facies (Fig. 5a), the specimens have a very low coercivity (coercive force lower than 8 mT) and saturation of ferrimagnetics seems to be attained in low fields (except for these low fields, the curves are superimposed for both increasing and decreasing field parts of the loop); this could indicate the presence of a weak amount of pyrrhotite or magnetite.

(b) In red facies (Fig. 5b), the hysteresis loop points out an incomplete saturation of the ferrimagnetics in higher fields, because substantial parts of the loop, for increasing and decreasing fields, are not superimposed; coercivity is high (coercive force of 30 to 60 mT), indicating that hematite is the main ferrimagnetic mineral.

In grey facies, Curie temperature has been studied, using a translation inductometer in a furnace. The sample was heated in air, in order to determine the mineral giving the low coercivity component (Fig. 6); the heating curve shows a weak susceptibility, with however a slight Hopkinson peak about 500°C before falling to zero at 590°C, the cooling curve shows an spectacular increase reaching a maximum at about 300°C. A second heating–cooling curve displays an increase of susceptibility, showing the instability of magnetic minerals of the sample at high temperature. The mineralogical transformation probably occurred from pyrite (Rolando, 1988) to magnetite (Bina et al., 1991) and possibly a low fraction of pyrrhotite at low temperature.

In red facies, demagnetization curves (Figs. 8b–8d) often show two decreases during thermal treatment (sometimes corresponding to a weak variation of magnetization direction—see Dourdou example of Fig. 9c): a weak decrease corresponding to magnetite Curie point, and the main decrease for higher temperature. The presence of

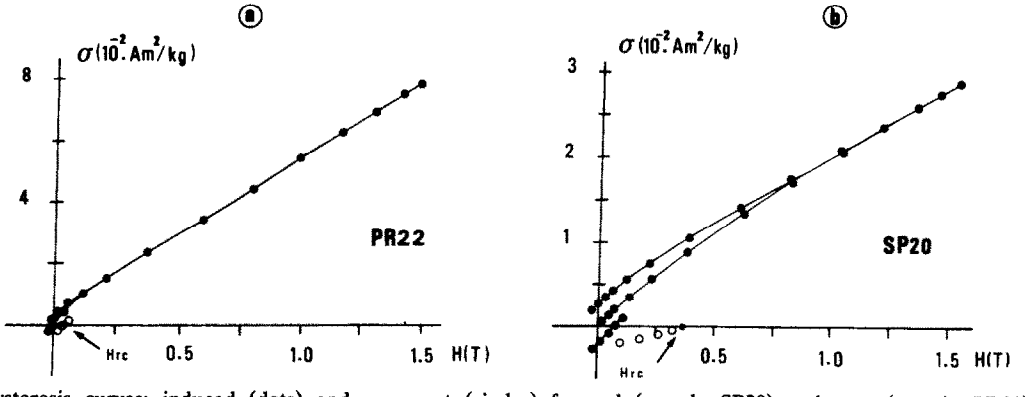


Fig. 5. Hysteresis curves: induced (dots) and remanent (circles) for red (sample SP20) and grey (sample PR22) pelites. *Hcr* = remanent coercive force; σ = specific magnetization; H = inducting field.

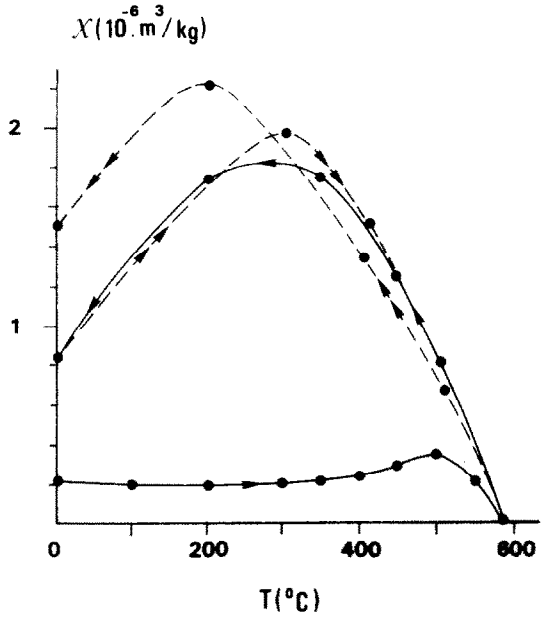


Fig. 6. Curie temperature curves in Saint-Rome 2a grey pelites (sample PR22): first (continuous line) and second (dotted line) heating–cooling cycles.

two different components, with different coercivities, was also observed in some samples of the red facies in the Lodève basin (Cogné et al., 1990). Hematite is therefore the main remanence carrier in this red facies, but a significant amount of magnetite is present in some samples, in particular in Saint-Rome 2b pelites and Dourdou sandstones.

On the other hand, in these grey and red pelites, transformation of pyrite to magnetite (Bina et al., 1991) occurs during the high-temperature treatments, and the last steps of the magnetization analysis do not allow the acquisition of usable results (even with the presence of hematite), because of a very strong increase in the magnetic viscosity; on the contrary, the pyrite amount in Dourdou sandstones (Rolando, 1988) is too low to produce a significant effect during thermal treatment.

Palaeomagnetic results

Saint-Rome 2a pelites

Thermal and alternating field treatments lead to similar results for this formation; different behaviours of the magnetization have been observed in both cases:

(a) All the samples from one site have well defined characteristic remanent magnetization

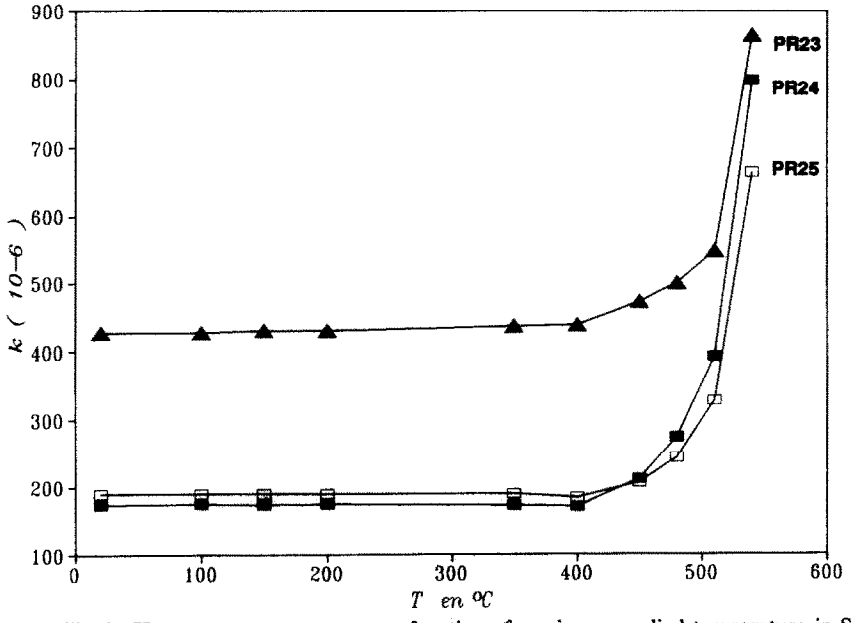


Fig. 7. Magnetic susceptibility in SI at room temperature as a function of maximum applied temperature in Saint-Rome 2a grey pelites.

(ChRM) along a N- to NNE-dipping direction (Table 1; Fig. 10a); a similar abnormal direction has been found in different sites of mineralized rocks of the Mesozoic cover of the southern French Massif Central (H. Rouvier and B. Henry, pers. commun., 1993). As barite veins have been observed near the studied sites, this direction clearly corresponds to a magnetic overprint likely related to the mineralisation location. Moreover, a Callovian–Bathonian age (about 165 Ma), in agreement with this ChRM direction, has been obtained on illite in an open vein of this basin (Bonhomme et al., 1980).

(b) Samples from other sites mostly yield a relatively well defined ChRM direction (Figs. 9a, 10a and Table 1), corresponding to the end of a progressive evolution along a part of the remagnetization circle during treatment (related to at least partial superimposition of different components); the low-temperature component has a similar orientation to the cluster of ChRM related to the overprint.

(c) Some other samples present poorly defined ChRM directions, because of the strong increase of magnetic viscosity related to thermal treatment at temperatures higher than 350°C or to the clearly incomplete separation of the two main components by alternating field treatment. The ChRM directions obtained are, however, close to those of the relatively well defined ChRM direction described in the previous paragraph.

As the dip is relatively constant and moderate in the northern border of the basin, the fold test is inapplicable. However, the application of a bivariate statistic (Le Goff, 1990; Le Goff et al., 1992) points out a surprising elliptic distribution of the obtained directions (k_x/k_y takes 5.2 and 5.0 values, respectively, before and after dip correction for the ChRM directions of 35 specimens). From the southern part of the same formation (Cogné et al., 1990), a similar high ellipticity has been obtained (using the directions from the three sites, k_x/k_y is 197.3—a very high value related to the distribution of the directions almost in a same plane—and 7.3, respectively, before and after dip correction); in this area, dip correction gives a strong decrease of ellipticity (which remains abnormally high, however), but also a significant negative Fisher fold test (α_{95} is 12° and 21°, respectively, before and after dip correction). These arguments indicate that the ChRM obtained does not represent the primary magnetization. This difference with primary magnetization could be explained either by a magnetization

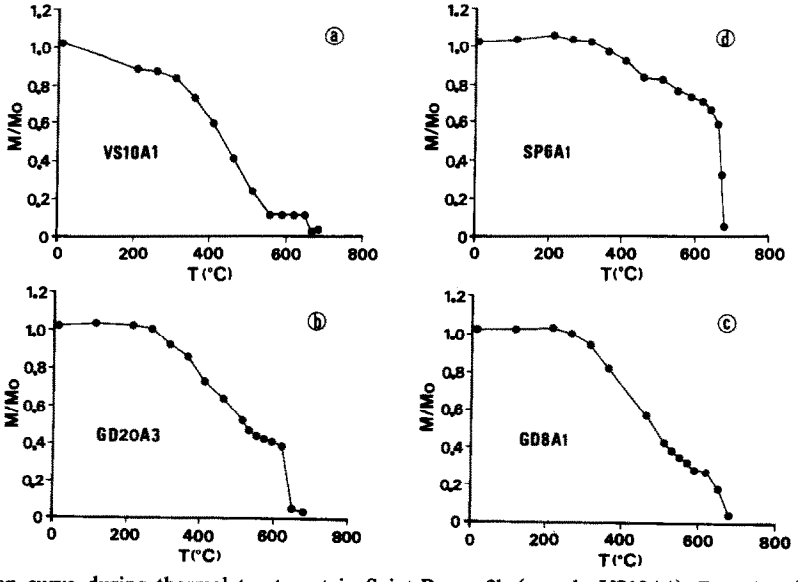


Fig. 8. Demagnetization curve during thermal treatment in Saint-Rome 2b (sample VS10A1), Dourdou (samples GD8A1 and GD20A3) and Saint-Pierre (sample SP6A1) formations.

acquisition during folding (which here could almost be synsedimentary—Legrand et al., 1991a, b), or by a superimposition of two magnetization components.

It can be noticed that the lengthening of the confidence ellipse of the ChRM directions (Le Goff, 1990; Le Goff et al., 1992), in the northern as well as in the southern part of the massif, almost corresponds to the vertical plane including the direction of the overprint likely related to mineralisation. The ChRM with a southern declination actually results from the superimposition of two components which cannot be totally separated by thermal or alternating field treatment, although the inclination of this ChRM seems rather close to some other Lower Permian European directions. Nevertheless, the plane containing the remagnetization circle of these samples is

TABLE 1

Saint-Affrique [a]

Site	N	Before dip correction				After dip correction				Lat. (°N)	Long. (°E)	K	A_{95} (°)
		D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)				
UPPER AUTUNIAN													
Saint-Rome 2a pelites (northern border)													
Site with only magnetic overprint													
a9	3	2.2	56.0	66	10.0	13.0	63.4	63	10.1				
Other sites													
a1	2	201.2	(18.7	3191	1.8)	200.0	(14.1	2929	1.8)				
a2	8	198.7	(15.1	66	6.1)	198.1	(9.6	66	6.1)				
a5	4	194.2	(−0.7	43	10.7)	194.4	(−6.4	43	10.7)				
a6	4	194.3	(9.3	15	18.2)	193.8	(1.1	15	18.2)				
a7	8	197.1	(9.6	34	8.5)	196.3	(9.3	34	8.5)				
a8	9	194.3	(13.7	18	10.9)	192.9	(13.5	18	10.9)				
Mean	35	196.5	(11.6	30	4.2)	195.7	(8.4	28	4.3)				
6 sites		196.6	(11.0	124	5.1)	195.9	(6.9	93	5.9)				
After clockwise rotation of 7.2°													
6 sites		203.8	(11.0	124	5.1)	203.1	(6.9	93	5.9)				
Saint-Rome 2a pelites (southern border—Cogné et al., 1990)													
p1	8	172	(3.5	43	8)	171	(17	43	8)				
p2	8	173	(10	123	5)	174	(9	123	5)				
p3	7	173	(−6	159	5)	171	(−9	159	5)				
3 sites		173	(2.5	101	8)	172	(6	37	14)				
After clockwise rotation of 7.2°													
3 sites		180.2	(2.5	101	8)	179.2	(6	37	14)				
SAXONIAN													
Saint-Rome 2b pelites													
b1	12	182.2	−5.5	122	3.7	182.0	−0.6	122	3.7				
b2	2	182.9	−7.1	130	8.7	182.7	−1.2	131	8.7				
b3	4	179.8	−2.3	250	4.4	179.8	3.5	250	4.4				
b6	6	181.5	0.9	192	4.1	181.6	6.8	192	4.1				
b7	4	184.2	−3.3	136	6.0	184.2	−1.7	136	6.0				
b8	9	183.1	−4.7	250	3.0	182.0	0.0	250	3.0				
b10	6	187.0	−5.3	121	5.2	185.5	−4.2	121	5.2				
Mean	43	182.3	−4.7	140	1.8	182.0	−0.5	130	1.8	47.3	179.9		
7 sites		182.8	−3.9	545	2.3	182.5	0.4	393	2.7	46.8	179.1		
7 sites		–	–	–	–	–	–	–	–	46.9	179.1	966	1.7
After clockwise rotation of 7.2°													
7 sites		190.0	−3.9	545	2.3	189.7	0.4	393	2.7				
7 sites		–	–	–	–	–	–	–	–	46.0	168.7	970	1.7

TABLE 1 (continued)

Saint-Affrique [a]

Site	N	Before dip correction				After dip correction				Lat. (°N)	Long. (°E)	K	A_{95} (°)
		D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)				
LATE SAXONIAN–LOWER THURINGIAN													
Dourdou sandstones													
d1	8	179.1	8.7	141	4.2	179.1	−8.7	123	4.5				
d2	8	184.2	17.9	62	6.3	184.0	−0.6	51	6.9				
d3	9	184.0	15.4	230	3.1	183.9	−5.2	236	3.0				
d4	8	185.4	20.2	106	4.8	185.8	−4.6	106	4.8				
d5	4	182.8	39.7	51	9.8	181.0	−5.5	49	10.0				
d7	6	186.4	12.9	411	2.8	186.0	−5.1	409	2.8				
d8	5	188.0	−10.0	162	4.9	184.4	−10.4	162	4.9				
d9	7	184.7	−0.2	124	4.8	184.8	4.6	125	4.7				
Mean	55	184.2	12.7	35	3.2	183.8	−4.3	86	2.0	49.1	177.0		
8 sites		184.4	13.1	30	9.1	182.6	−4.4	190	3.6	49.2	178.8		
8 sites		–	–	–	–	–	–	–	–	49.2	177.3	587	2.0
After clockwise rotation of 7.2°													
8 sites		191.6	13.1	30	9.1	189.8	−4.4	190	3.6				
8 sites		–	–	–	–	–	–	–	–	48.2	166.4	585	2.0
MIDDLE THURINGIAN													
Saint-Pierre pelites													
1	10	192.2	0.7	121	4.0	192.3	−7.9	116	4.1				
s3	2	189.3	−9.9	–	–	189.2	−11.9	–	–				
s4	4	188.6	−1.1	162	5.5	188.6	−0.8	162	5.5				
s5	3	190.7	−12.9	50	11.5	191.4	−17.1	50	11.5				
s6	9	193.1	−5.5	64	5.8	193.4	−7.3	64	5.8				
s7	5	196.0	−0.7	81	6.9	196.0	−7.7	81	6.9				
s8	3	192.6	5.8	108	7.8	192.4	2.8	108	7.8				
s9	7	195.6	−10.4	42	8.2	195.7	−17.4	42	8.2				
Mean	43	192.8	−3.7	61	2.7	193.0	−8.6	57	2.8	49.7	162.5		
8 sites		192.3	−4.3	134	4.3	192.4	−8.4	114	4.6	49.8	163.4		
8 sites		–	–	–	–	–	–	–	–	49.8	163.4	323	2.8
After clockwise rotation of 7.2°													
8 sites		199.5	−4.3	134	4.3	199.6	−8.4	114	4.6				
8 sites		–	–	–	–	–	–	–	–	47.7	153.1	322	2.8

[a] For each cluster of ChRM in each site, number of specimens (N), mean direction (D, I) before and after dip correction. Mean values for each cluster and paleomagnetic pole (lat., long.). Corresponding Fisher's (1953) parameters (k, α_{95}, K, A_{95}). Data between parentheses: only declination is significant (see text).

almost vertical, with a very low scattering of the ChRM declination; the declination value obtained can, therefore, be assigned to the declination of the primary magnetization, the effect of the secondary component being limited to a scattering of the ChRM inclination. The declination from these pelites is, therefore, usable for the study of local or regional rotations.

In fact, in the Saint-Affrique basin, this declination is very well defined ($\alpha_{95} = 2°$) using the bivariate statistic, in the southern (Cogné et al., 1990) as well in the northern borders; the large difference in declination (24°) between these two areas cannot be explained by uncertainty of the data, but clearly implies the occurrence of a tectonic event.

Saint-Rome 2b pelites

The analysis of the Zijderveld diagrams (Fig. 9b) points out two different components. The first one lies to the present earth magnetic field,

and is totally removed after heating at 250°C. The second component has most of the time a linear demagnetization path towards the origin for temperatures from 300°–350°C to 580°C, and even sometimes to 640°C; thus it is probably carried by both magnetite and hematite. At higher temperatures, several samples present very important disturbances (in particular, very strong increase of the magnetic viscosity) related to mineralogical variation due to heating, and the last steps of the analysis become then unusable.

For the ChRM distribution obtained on 43 specimens (Fig. 10b), dip correction does not significantly change the Fisher parameter, but gives a decrease from 1.7 to 1.3 for the ratio k_x/k_y of the ellipticity (Le Goff, 1990; Le Goff et al., 1992), which is very low after this correction. Moreover, the lengthening of the distribution ellipse is perpendicular to the fold axis before dip correction; the bivariate statistic yields a positive fold test here, and indicates the absence of superimposed magnetization in the ChRM. The grey

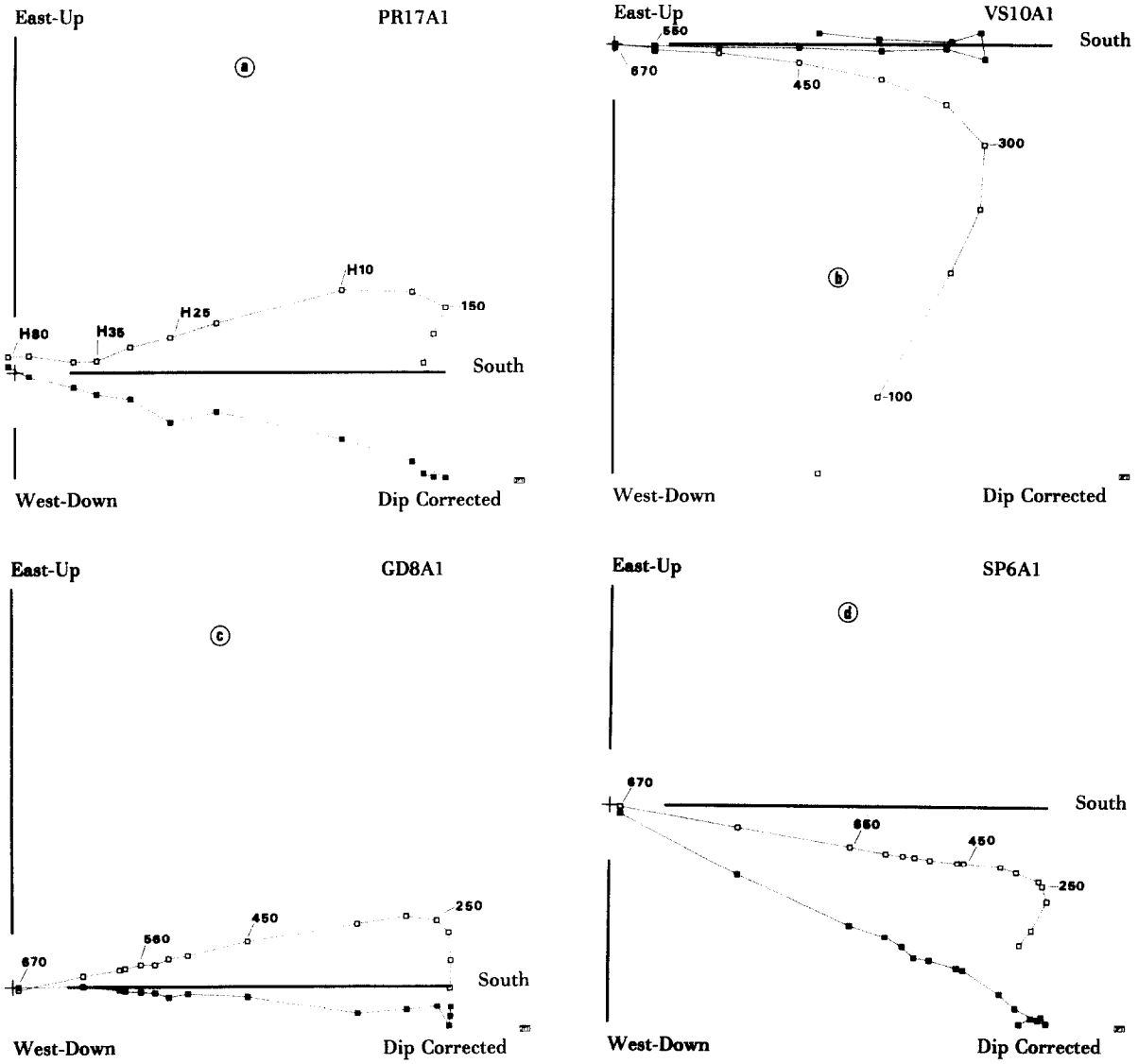


Fig. 9. Orthogonal vector plot with heating value in degrees Celsius and alternating field (H) in mT (black symbols, horizontal plane; white symbols, vertical plane) in Saint-Rome 2a (sample PR17A1), in Saint-Rome 2b (sample VS10A1), Dourdou (samples GD8A1) and Saint-Pierre (sample SP6A1) formations

and red pelites give, then, a very well defined mean direction of the ChRM (Table 1) The pre-folding age of this ChRM shows here that it corresponds to primary magnetization, the folding having begun in this area during deposition (Legrand et al., 1991a,b).

Dourdou sandstones

In some samples, two slightly different components can be separated by thermal treatment (Fig. 9c); they are probably related to different magnetic carriers (magnetite and hematite), but their mean directions at the scale of the whole formation are not significantly distinct (in most cases, the directions of the low- or high-temperature components show the same values of inclination and declination. However, differences in inclination, in the upwards or downwards direction can be observed, but in average, they are not significant); these components were likely acquired at different stages of the diagenesis, and thus reflected the magnetic field direction during this period. When these two components are present, the highest blocking temperature component (carried by hematite) was chosen because of its presence in all the samples.

In these levels, sampling was carried out in areas with different dip; a fold test has been applied to ChRM of 55 specimens. Fisher parameter k varies from 35 to 86, and the k_x/k_y ratio of ellipticity from 6.2 to 2.0; here also, the lengthening of the ellipse is perpendicular to the fold axis before dip correction, with low ellipticity after dip correction (Fig. 11). This positive fold test shows that the obtained ChRM (Table 1) has been acquired before folding and, therefore, is definitely a primary magnetization.

Saint-Pierre pelites

As for Dourdou red sandstones, the ChRM of Saint-Pierre pelites (Fig. 10d and Table 1) is carried, at least for the main part, by hematite,

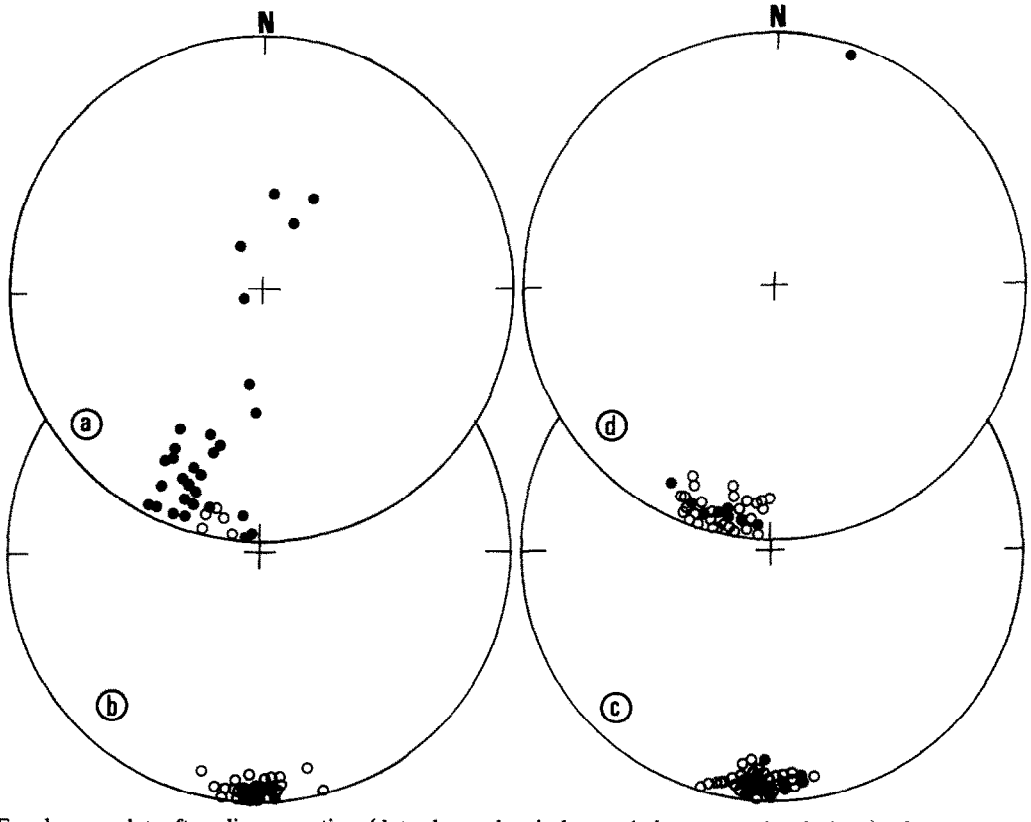


Fig. 10. Equal area plot after dip correction (dots, lower hemisphere; circles, upper hemisphere) of ChRM vectors from Saint-Rome 2a (a) and 2b (b), Dourdou (c) and Saint-Pierre (d) formations.

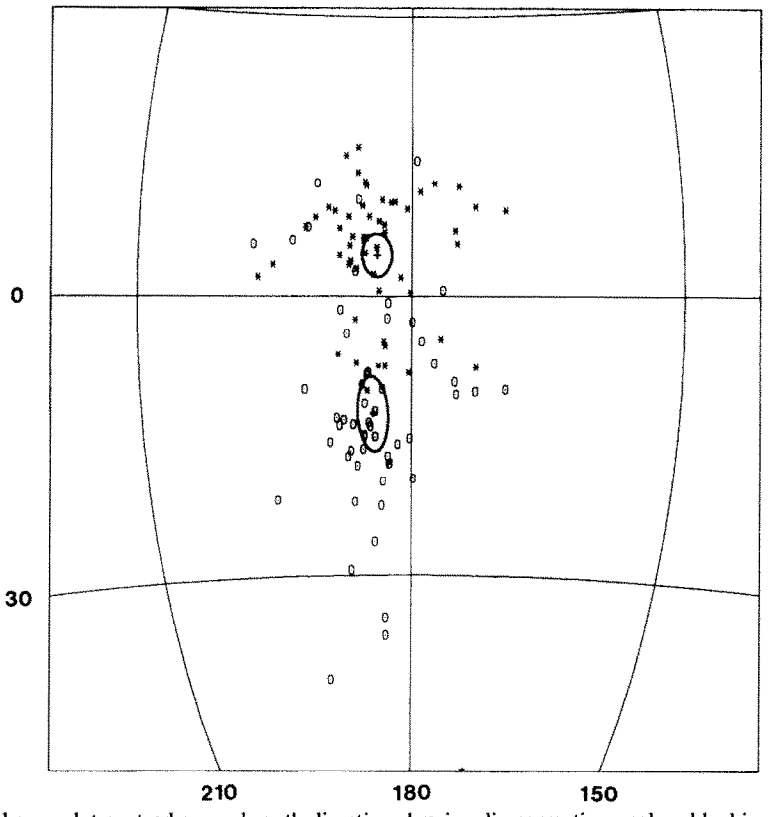


Fig. 11. Equatorial equal area plot centred around south direction showing dip correction analyzed by bivariate statistics (Le Goff, 1990; Le Goff et al., 1992): ChRM of Dourdou sandstones before (circles) and after (asterisks) dip correction.

with a linear path on the demagnetization diagram (Fig. 9d), high unblocking temperature (670°C), and high coercivity. The dip variations being very low, the fold test is insignificant (the k_x/k_y ratio of ellipticity is moderate, about 2, for the distribution of ChRM of 43 specimens).

One specimen from a handsample of the upper part of the Saint-Pierre formation has a ChRM with normal polarity (two others present components of reversed polarity for the lowest temperature and evolution towards normal polarity during analysis at the highest temperatures); the normal and reversed direction of Saint-Pierre pelites are perfectly antiparallel, pointing out the lack of overprint in this formation, and indicating a likely primary origin of the ChRM. The presence of three normal directions in specimens from Saint-Pierre pelites (only in the upper part of the series) leads to an age corresponding to the end of the Kiaman interval, probably middle Thuringian (the first known normal polarity during the Upper Permian is of late Murghabian age —Kotlyar et al., 1984); this age is in agreement with K–Ar dating (Bonhomme et al., 1980).

Discussion

Comparison with palaeomagnetic results from the Lodève basin (Table 2)

Figure 12 presents the comparison of the declinations measured in similar formations of the Saint-Affrique and Lodève (Henry, 1989, modified from Mérabet and Guillaume, 1988) basins: Saint-Rome 2a and Tuileries-Loiras L2 (Autunian), Saint-Rome 2b and Tuileries-Loiras L3 (Saxonian) and Saint-Pierre and Salagou (Thuringian). Except for the Autunian, the Saint-Affrique palaeomagnetic directions from different sites within each formation are similar in the entire basin, and can be compared to the Lodève directions. A systematic angular deviation

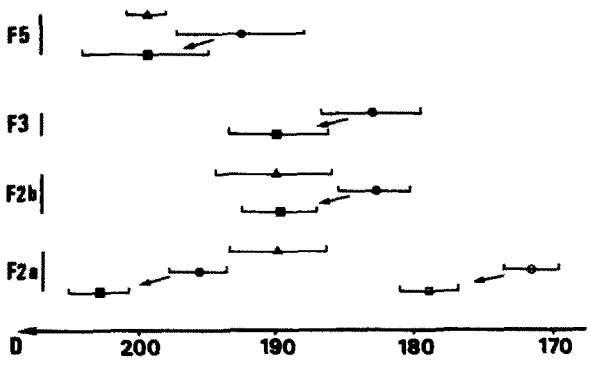


Fig. 12. ChRM declination (recalculated for Saint-Affrique coordinates) according to the age in the basins of Lodève (triangles) and of Saint-Affrique (dots); they are compared to those of Saint-Affrique, after clockwise rotation of 7.2° (squares): white symbols (Cogné et al., 1990) and black symbols (this study).

(Autunian excepted) exists between the results from the two basins, pointing out an anticlockwise rotation with a mean of 7.2° (7.7° for the Saxonian and 6.8° for the Thuringian), after the deposition of Saint-Pierre pelites, of the Saint-Affrique basin relative to the Lodève basin.

To compare the results of the different formations from the two basins, a clockwise rotation of this angle (7.2°) has then been applied to the Saint-Affrique directions (Fig. 12). Similarities are found between the Saxonian and Thuringian directions, but southern and northern parts of the Saint-Affrique basin have respectively undergone, before the Saxonian, an anticlockwise rotation of about 11°, and a clockwise rotation of 13°, relative to the Lodève basin.

Comparison with other palaeomagnetic data from stable Europe

The question is now to determine the rotation of these two basins relative to stable Europe. Palaeomagnetic poles (Fig. 13) of the Saint-Affrique basin (before and after the clockwise rotation of 7.2°) and the Lodève basin (Mérabet and Guillaume, 1988; Henry, 1989) have been situated in relation to the Apparent Polar Wander Path (APWP) for stable Europe. They were obtained by applying bivariate statistics (Le Goff et al., 1992) on the compilation of results, except, obviously, those from Lodève (Van der Voo, 1990).

The Lodève and Saint-Affrique poles (corrected by rotation) appear to follow this APWP for stable Europe. However, as suggested by geological arguments (Van Den Driessche and Brun, 1989), the Saint-Affrique basin was affected by rotations, whereas Lodève basin was not.

Stratigraphical and structural implications

Permian opening of the basin

The general context of the opening of the Saint-Affrique basin is a NE–SW extension

TABLE 2

Lodève [a]

Site	N	Before dip correction				After dip correction				Lat. (°N)	Long. (°E)	K	A_{95} (°)
		D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)				
UPPER AUTUNIAN													
5 sites		190.6	8.1	411	3.1	190.0	−2.0	327	3.5				
5 sites		–	–	–	–	–	–	–	–	41.3	169.3	464	2.9
SAXONIAN													
3 sites		190.1	3.5	351	4.3	190.2	−8.3	351	4.3				
3 sites		–	–	–	–	–	–	–	–	43.7	169.5	666	3.1
THURINGIAN													
6 sites		198.9	−0.3	1276	1.6	199.2	−11.8	1451	1.5				
6 sites		–	–	–	–	–	–	–	–	48.6	153.5	2096	1.2

[a] For each cluster of ChRM in each site, number of specimens (N), mean direction (D, I) before and after dip correction. Mean values for each cluster and paleomagnetic pole (lat., long.). Corresponding Fisher's (1953) parameters (k, α_{95}, K, A_{95}) (Mérabet and Guillaume, 1988; Henry, 1989).

(Santouil, 1980) and therefore without rotation, except between the Autunian and Saxonian (at least for the southern and northern borders of the basin).

Concerning the rotation between the Autunian and Saxonian, the assumption of a single chisel opening of a graben along normal plane faults appears hardly acceptable, because of the importance of the rotations (13° clockwise and 11° anticlockwise rotations in the northern and southern border, respectively), implying very strong extension towards the west (there is no clear evidence of such an extension in the Albigeois thrust sheets).

The local anticlockwise rotation on the southern border could be related to sinistral movements along the border faults, which might be related to the l'Espinouse structure and transfer faults during the NE–SW extension. The local character of this rotation does not yield any conclusion about the geometry of these faults (other Permian formations from the neighbouring area do not show a similar rotation): although the rotation does correspond to the emergence of listric faults under the Saint-Affrique basin (Van Den Driessche and Brun, 1989), the entire detached northern block located over these faults is not affected. In contrast, the northern border was affected by clockwise rotation, of which the origin could also be local phenomena related to dextral movements along E–W faults. In fact, in a context of a NE–SW extension, the geometry of the Plaisance–l'Espinouse–Cévennes faults might provoke anticlockwise rotation of the whole area around the l'Espinouse fault and thus, the displacement of the Saint-Affrique basin as a whole toward the west along the E–W faults.

Fold axes of phase 2 of the Hercynian orogeny have mainly NNW–SSE to E–W orientations north (southern Lévezou—Collomb, 1970; Nicol-

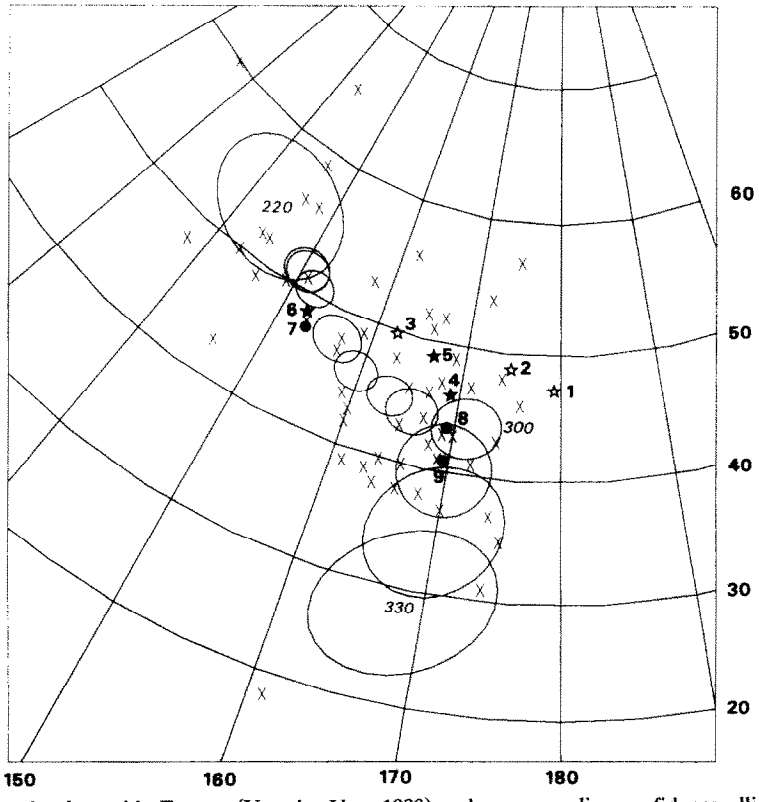


Fig. 13. Palaeomagnetic poles for stable Europe (Van der Voo, 1990) and corresponding confidence ellipses at 10 m.y. intervals (weighting by number of sites; bivariate statistics—Le Goff, 1990; Le Goff et al., 1992) for APWP, for Saint-Affrique, before (white stars, 1–3) and after (black stars, 4–6) clockwise rotation of 7.2°, and for Lodève (dots, 7–9): upper Autunian (9), Saxonian (1, 4 and 8), late Saxonian–lower Thuringian (2 and 5) and middle Thuringian (3, 6 and 7).

let, 1978) and west (Albigeois thrust sheets—Delpont, 1973; Bles et al., 1982; Collomb and Meyzindi, 1991) of the Saint-Affrique basin, and E–W to NE–SW south of the basin (Gèze, 1949; Arthaud, 1970; Monts de Lacaune: Donnot and Guérangé, 1978; Southeastern border of the Sidobre: Béziat, 1973; and axial zone of the Montagne Noire: Bard and Loueyit, 1978). Similarly, some old faults have a mainly E–W orientation in the western part of the basin (Brousse-Broquies Stephanian graben) and a ENE–WSW orientation in the southeastern border (the Hercynian Monts de Lacaune); these fault orientations have also been observed in the Permian levels (and in the Mesozoic cover east of the Saint-Affrique basin), mainly E–W in the northern and central parts, and ENE–WSW in the southern part, probably because of a reactivation of old structures. The angular difference between these different directions is of the same direction and of the same order of amplitude as the difference between Autunian palaeomagnetic directions in the northern and southern borders of the Saint-Affrique basin. This seems to indicate that differential rotations (reactivation of the thrust sheets of the southern Albigeois?) were produced during an extension of the basin between Autunian and Saxonian times and, therefore, suggests the existence of a major structure under the Permian and Mesozoic covers.

Age of the Saint-Affrique formations

An upper Autunian palynological age is well established by flora Lo1 (Doubinger et al., 1987) found in the F1 and Saint-Rome 2a formations (Rolando et al., 1989). Flora Lo3, from Dourdou sandstones, indicates a lower Thuringian age (Doubinger et al., 1987; Rolando, 1988), or possibly a late Saxonian age (Schaarschmidt, 1980; Doubinger et al., 1987). The close mean directions of ChRM of the Saint-Rome 2b and Dourdou sediments show that the difference in age between these two formations is relatively small; if a lower Saxonian age is attributed to the Saint-Rome 2b formation by comparison (Rolando et al., 1989) with the L3 level of Lodève series well dated by flora Lo2 (Doubinger et al., 1987), this favours a late Saxonian to early Thuringian age for the Dourdou sandstones. On the other hand, as indicated before, palaeomagnetic data make it possible to attribute Saint-Pierre pelites to a middle Thuringian age. Events responsible for the main unconformity in the Lodève basin (under the Rabejac formation) and Saint-Affrique basin (under the Belmont conglomerates) occurred therefore during the lower Thuringian, or at the Saxonian–Thuringian boundary.

Late Thuringian or more recent rotation

Contrary to the Autunian rotations, the anticlockwise post-middle Thuringian rotation has uniformly affected the entire Saint-Affrique basin. The uniform character of this rotation suggests a movement affecting the cover and, at the least, a large part of the basement. The extent of the rotated area cannot be presently specified, except that the Lodève basin is not affected.

The age of this rotation is at present not yet determined. The absence of deposits between the middle Thuringian and Middle Triassic does not allow to obtain information about this period. The lack of obvious traces of such a large rotation in the Mesozoic cover might be an argument in favour of occurrence of this event during the Late Permian–Lower Triassic; there was a similar weak rotation during the Palatine phase in the Montagne Noire (Gèze, 1949); this could also correspond to the N–S extension giving some dextral movements along deep structures (Legrand, 1990). Moreover, palaeomagnetic data obtained in the Bathonian basalt from Les Vignes, in the Mesozoic cover, do not show anticlockwise rotation (Fabre, 1986); extension towards the east of the rotated area being unknown, this argument does not yield a definite conclusion, and the assumption of a more recent rotation, for example, during a Callovian event (Bonhomme et al., 1980), or in relation with the rotation of Iberia during the Early Cretaceous (Galdeano et al., 1989) or Late Cretaceous (Platzman and Lowrie, 1992) or even during the Oligocene extension, cannot be excluded.

The existence of a major structure between the Lodève and Saint-Affrique basins without traces

of major post-middle Thuringian compression or extension in this area, and the large size of the zone affected by the rotation, seem to be an argument in favour of movement along a major listric structure at great depth under the Saint-Affrique basin (Van Den Driessche and Brun, 1989) after Hercynian time.

Conclusions

Palaeomagnetic analyses carried out in the Permian Saint-Affrique basin lead to assumptions of a late Saxonian to early Thuringian age for Dourdou sandstones and of a middle Thuringian age for Saint-Pierre pelites. Contrary to the Lodève basin, the Saint-Affrique area has undergone, after the middle Thuringian (Palatine phase or more recent tectonic event), an anticlockwise rotation on the order of 7° relative to stable Europe. The opening of the basin occurred without rotation from the Saxonian to middle Thuringian, but large rotations, respectively clockwise and anticlockwise in the northern and southern border of the basin, earlier affected the Saint-Affrique area between the upper Autunian and Saxonian. These results show the different stages of evolution of this area: an extension with large rotations up to the Saxonian, a period without rotation up to at least the middle Thuringian, and a later phase of regional rotation.

Taking into account the importance of the structural implications of these results, it is of major importance to determine the areas affected by the pre-Saxonian and post-Thuringian rotations, as well as the age of this last rotation.

In order to obtain information on the pre-Saxonian rotation, a palaeomagnetic study of other Autunian formations, and in particular those of the Rodez basin (Bourges, 1987) is likely to be the only feasible approach. Palaeomagnetic studies of other Permian and Mesozoic formations should give information on the area affected and on the age of the post-middle Thuringian rotation. These studies will therefore be of major importance for a better understanding of the structural evolution of the French Massif Central, even before Autunian time.

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